Straight Line: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Quadrilaterals and circle facts

Formulas

Quadrilateral: $\frac{1}{2}d_1 d_2 \sin\theta$
Parallelogram: $ab\sin\theta = p_1 p_2 \operatorname{cosec}\theta$
Trapezium: $\frac{1}{2}(a+b)h$; median $\frac{a+b}{2}$

KEY POINTS

- Parallelogram: opposite sides equal and parallel, opposite angles equal, adjacent angles supplementary, diagonals bisect each other, $2(a^2+b^2) = d_1^2 + d_2^2$
- Rhombus: all sides equal, diagonals perpendicular and bisect the angles, $p_1 = p_2$; rectangle: angle $90°$, diagonals equal; square: both
- Trapezium: exactly one pair of parallel sides; isosceles $\iff$ base angles equal (legs equal, diagonals equal)
- Cyclic quadrilateral: $A+C = B+D = \pi$; Ptolemy $AB\cdot CD + AD\cdot BC = AC\cdot BD$; chords $PA\cdot PB = PC\cdot PD = PT^2$

STEPS

1. Vertices given, test parallelogram: mid-points of the diagonals coincide
2. Sides given, test rhombus: two parallel pairs and equal distances $p_1 = p_2$

2. Coordinates, polar form, distance

Formulas

$$PQ = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

$$x = r\cos\theta,\ \ y = r\sin\theta,\ \ r = \sqrt{x^2+y^2}$$

Rotation by α: $(x\cos\alpha - y\sin\alpha,\ \ x\sin\alpha + y\cos\alpha)$

KEY POINTS

- On x-axis $(h, 0)$, on y-axis $(0, k)$; horizontal line: same y, vertical line: same x
- θ: I α, II $\pi-\alpha$, III $-(\pi-\alpha)$, IV $-\alpha$, with $\alpha = \tan^{-1}\left|\frac{y}{x}\right|$
- Nature of triangle (c largest): $a^2+b^2 >, =, < c^2$: acute, right, obtuse
- Angles from vertices: $\cos A = \dfrac{b^2+c^2-a^2}{2bc}$

3. Section formula and harmonic conjugate

Formulas

Internal: $\left(\dfrac{mx_2+nx_1}{m+n}, \dfrac{my_2+ny_1}{m+n}\right)$
External: $\left(\dfrac{mx_2-nx_1}{m-n}, \dfrac{my_2-ny_1}{m-n}\right)$

KEY POINTS

- Trisection points: ratios $1:2$ and $2:1$
- Ratio in which a line/axis divides AB: take $\lambda:1$, use one coordinate; $\lambda<0$ means external
- Harmonic conjugates P, Q: same ratio internally and externally; AP, AB, AQ in H.P.; $MP\cdot MQ = MB^2$

4. Centres of a triangle

Formulas

$$G = \frac{A+B+C}{3}$$

$I = \dfrac{aA+bB+cC}{a+b+c}$; I_1: $a \to -a$
O: weights $\sin 2A, \sin 2B, \sin 2C$; H: weights $\tan A, \tan B, \tan C$

Euler line: $G = \dfrac{H + 2O}{3}$

KEY POINTS

- Right triangle: H = right-angle vertex, O = mid-point of hypotenuse
- Equilateral: all four coincide; isosceles: all four collinear
- Circumcentre at origin $\Rightarrow H = (x_1 + x_2 + x_3,\ y_1 + y_2 + y_3)$
- Orthocentre of BCH is A; image of H in a side lies on the circumcircle
- 4th vertex of parallelogram $ABCD$: $D = A + C - B$
- Equal areas $PAB = PBC = PCA \Rightarrow P = G$

STEPS

1. O: solve $OA^2 = OB^2$, $OB^2 = OC^2$
2. H: intersect two altitudes
3. Mid-points of sides given: vertices $= D + E - F$ etc.

5. Area and collinearity

Formulas

$$\Delta = \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Polygon: $\frac{1}{2}|\sum(x_i y_{i+1} - x_{i+1} y_i)|$

Three lines: $\Delta = \dfrac{(\det)^2}{2|C_1C_2C_3|}$

KEY POINTS

- Collinear $\iff \Delta = 0 \iff$ equal slopes
- Polygon vertices must be taken in order (clockwise or anticlockwise)
- Triangle on mid-points has $\frac{1}{4}$ the area; triangle with axes $\frac{1}{2}|ab|$

Watch out

Area is a modulus: '$\Delta = k$' gives two cases $\pm k$

6. Locus and standard loci

STEPS

1. Take $P(h, k)$, write the condition, eliminate parameters, replace h, k by x, y
2. Parametric $h = f(t),\ k = g(t)$: eliminate t

KEY POINTS

- $PA = PB$: perpendicular bisector; $PA = kPB$ $(k \neq 1)$: circle
- $\angle APB = 90°$: circle on AB as diameter; $\angle APB = \alpha$: arcs, $R = \dfrac{AB}{2\sin\alpha}$
- $PA + PB = k > AB$: ellipse; $|PA - PB| = k < AB$: hyperbola; $= AB$: segment / two rays
- Parabola $PS = PM$ (focus, directrix); ellipse $PS_1 + PS_2 = 2a$; hyperbola $|PS_1 - PS_2| = 2a$

7. Slope, intercepts, forms of a line

Formulas

$$m = \tan\theta = \frac{y_2 - y_1}{x_2 - x_1} = -\frac{a}{b}$$

$y - y_1 = m(x - x_1)$, $y = mx + c$

$\dfrac{x}{a} + \dfrac{y}{b} = 1$, $x\cos\alpha + y\sin\alpha = p$

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

KEY POINTS

- Inclination $0 \le \theta < \pi$; horizontal $m = 0$, vertical m undefined; parallel $m_1 = m_2$
- Intercepts of $ax + by + c = 0$: $-\frac{c}{a}$, $-\frac{c}{b}$; equal intercepts $m = -1$; equal in magnitude or equally inclined $m = \pm 1$
- Point on a line: $(t,\ mt + c)$; first-degree equation $\iff$ line
- Normal form: divide by $\pm\sqrt{a^2 + b^2}$ so that $p > 0$
- Mid-point of intercept $(\frac{a}{2}, \frac{b}{2})$; foot from O (h, k): line $hx + ky = h^2 + k^2$
- Line through (a, b) cutting positive axes: least area $2ab$, least $OA + OB = (\sqrt{a} + \sqrt{b})^2$

Watch out

$y - y_1 = m(x - x_1)$ and $y = mx + c$ miss the vertical line: check $x = x_1$

8. Parametric form

Formulas

$$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$$

Point: $(x_1 + r\cos\theta,\ y_1 + r\sin\theta)$

KEY POINTS

- Use for distances along a line: intercept length, distance measured along / parallel to a line, $PA \cdot PB$
- Sign of r tells the side of (x_1, y_1)
- Equilateral triangle on AB: third vertex at $\frac{\sqrt{3}}{2}AB$ from mid-point, along the perpendicular

9. Angle between lines, foot and image

Formulas

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

Foot: $\dfrac{h - x_1}{a} = \dfrac{k - y_1}{b} = -\dfrac{ax_1 + by_1 + c}{a^2 + b^2}$

Image: same with $-\dfrac{2(ax_1 + by_1 + c)}{a^2 + b^2}$

KEY POINTS

- Parallel $ax + by + \lambda = 0$; perpendicular $bx - ay + \lambda = 0$; $m_1 m_2 = -1$
- Equally inclined to x-axis: $m_1 + m_2 = 0$ (bisectors horizontal / vertical); $m_1 m_2 = 1$: bisectors of slopes ± 1
- Bisector inclinations $\frac{\theta_1 + \theta_2}{2}$, $\frac{\theta_1 + \theta_2}{2} + \frac{\pi}{2}$
- Line at angle α with slope m_1: two slopes $\dfrac{m_1 \pm \tan\alpha}{1 \mp m_1 \tan\alpha}$
- Reflected ray passes through the image of a point of the incident ray
- Triangle angles from sides: angle between each pair; check acute/obtuse at vertices

Watch out

A lost root in the slope equation = a vertical line

10. Distances, parallel lines, areas

Formulas

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$d = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$$

Parallelogram: $\dfrac{|(c_1 - d_1)(c_2 - d_2)|}{|a_1 b_2 - a_2 b_1|} = \dfrac{p_1 p_2}{\sin\theta}$

Equilateral: $\Delta = \dfrac{h^2}{\sqrt{3}}$; right isosceles: $\Delta = h^2$

Watch out

Make x, y coefficients identical before using the parallel-line formulas

11. Position of points, regions, concurrency

KEY POINTS

- Same side $\iff L(P)\,L(Q) > 0$; opposite $\iff L(P)\,L(Q) < 0$
- Region $ax + by + c > 0$: test the origin; $b > 0$: above the line
- Inside $\triangle ABC$: P and each vertex on the same side of the opposite side
- $|x| + |y| \le a$ and similar: split into quadrants / regions
- Concurrent: $\det[a_i\ b_i\ c_i] = 0$ (lines not parallel)
- No triangle: two lines parallel or all three concurrent

12. Images

KEY POINTS

- (a, b): x-axis $(a, -b)$, y-axis $(-a, b)$, origin $(-a, -b)$, $y = x$ (b, a), $y = -x$ $(-b, -a)$
- Curve: replace (x, y) by the image of (x, y), e.g. $y = x$: $f(y, x) = 0$
- Image of fixed P in lines through fixed Q: circle, centre Q, radius QP
- Image of A in perpendicular bisector of AB is B; image of B in bisector of $\angle A$ lies on AC
- Image of a line: through (line $\cap$ mirror) and the image of one of its points

13. Family of lines

Formulas

$$L_1 + \lambda L_2 = 0$$

KEY POINTS

- Type 1: lines through $L_1 = 0 = L_2$; find λ from the extra condition
- λ impossible $\Rightarrow$ answer is $L_2 = 0$
- Type 2: parameter linear (or linear relation in a, b, c) $\Rightarrow$ fixed point
- Farthest line from P through A: perpendicular to PA, distance PA
- Diagonals of parallelogram $u = 0, k$; $v = 0, l$: $\dfrac{u}{k} + \dfrac{v}{l} = 1$, $\dfrac{u}{k} = \dfrac{v}{l}$

14. Shifting and rotation of axes

Formulas

Shift to (h, k): $x = X + h,\ y = Y + k$
Rotation θ: $x = X\cos\theta - Y\sin\theta,\ y = X\sin\theta + Y\cos\theta$

$$X = x\cos\theta + y\sin\theta,\ Y = -x\sin\theta + y\cos\theta$$

KEY POINTS

- New equation: substitute old x, y in terms of X, Y
- Clockwise rotation: $\theta \to -\theta$
- To remove first-degree terms shift the origin to the centre

15. Angle bisectors

Formulas

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

KEY POINTS

- Acute bisector: tan(angle with L_1) < 1, or a point of L_1 is nearer to it
- Containing P: make $L_1(P), L_2(P)$ same sign, take '+'

- $c_1, c_2 > 0$: $a_1a_2 + b_1b_2 < 0 \Rightarrow$ origin bisector acute; > 0 obtuse
- Triangle, vertices known: $BD : DC = c : b$; sides known: sign method ($L_{AB}(C) > 0$, $L_{AC}(B) > 0$, '+')
- Internal bisector separates B and C; a line equally inclined to two lines is parallel to a bisector

16. Shortest and longest paths

KEY POINTS

- Min $PA + PB$: same side, reflect B; opposite sides, join AB
- Max $|PA - PB|$: same side, join AB (max AB); opposite sides, reflect B (max AB')
- Min $|PA - PB| = 0$: on the perpendicular bisector of AB

17. Pair of straight lines

Formulas

$ax^2 + 2hxy + by^2 = 0$: $m_1 + m_2 = -\dfrac{2h}{b}$, $m_1m_2 = \dfrac{a}{b}$

$$\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|$$

Bisectors: $\dfrac{x^2 - y^2}{a - b} = \dfrac{xy}{h}$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

KEY POINTS

- $h^2 > ab$ real distinct, $= ab$ coincident / parallel, $< ab$ imaginary
- Perpendicular $a + b = 0$; $m_1 + m_2 = 0 \iff h = 0$; $m_1m_2 = 1 \iff a = b$
- Slopes from $bm^2 + 2hm + a = 0$ (2nd-degree part only)
- Homogeneous of degree n: n lines through O; put $y = mx$
- Point of intersection / centre: $ax + hy + g = 0$, $hx + by + f = 0$
- Parallel pair: distance $2\sqrt{\dfrac{g^2 - ac}{a(a + b)}}$

STEPS

1. Factorise: split the 2nd-degree part, then compare x, y, constant terms
2. Bisectors of a general pair: shift the origin to the point of intersection first

18. Homogenisation

STEPS

1. Line $lx + my = n$: write $1 = \dfrac{lx + my}{n}$
2. Multiply 1st-degree terms by it, the constant by its square
3. Result = pair OA, OB: right angle $a + b = 0$; equally inclined $h = 0$

KEY POINTS

- Chords subtending 90° at O of a fixed conic pass through a fixed point

19. Common mistakes

STEPS

1. Forgetting the vertical line $x = x_1$ in slope forms
2. Using the parallel-line formula without equal x, y coefficients
3. Taking only one sign of a modulus (area, distance)
4. Wrong order of polygon vertices
5. Mixing internal and external bisectors of a triangle
6. Not checking $\lambda \to \infty$ ($L_2 = 0$) in the family $L_1 + \lambda L_2 = 0$
7. Using $\tan\theta$ formula of a pair without checking $h^2 \geq ab$